

Madhav Singh Bhanot

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SUMMARY

Full-Stack Software Engineer specializing in **AI/LLM integration**, scalable web architectures, and production-grade delivery. Experienced at **Innate AI** and **KrishLabs** building **RAG pipelines**, REST APIs, and streaming architectures with **React**, **Next.js**, **Node.js**, **Express**, **PostgreSQL**, and **MongoDB**. Proficient in OpenAI & Anthropic APIs, prompt engineering, and AWS cloud deployment. Hackathon award winner adept at turning complex requirements into high-impact products.

EDUCATION

Bharati Vidyapeeth Deemed University | Pune, India |

Graduated 2026

Bachelor of Technology in Computer Science and Engineering | **CGPA: 8.32/10**

SKILLS

Languages: JavaScript (ES6+), TypeScript, Java, SQL, HTML5, CSS3

Frontend: React.js, Next.js, React Native (Expo), Redux, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, WebSockets, Socket.IO, RESTful APIs, Prisma ORM, Mongoose, JWT

Databases & Storage: PostgreSQL, MongoDB, Redis, Supabase, Firebase Firestore

Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD Pipelines, Git, Postman, Jest, Vite

Data & Workflow: Apache Airflow, Celery, ETL Pipelines, Web Scraping

AI & LLM Engineering: OpenAI API, Anthropic API, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search, LLM Workflows

EXPERIENCE

Innate AI | React Developer (Internship) | Helsinki, Finland Remote |

Jul 2025 – Oct 2025

- **Engineered mission-critical LLM dashboards:** Architected high-throughput React + TypeScript dashboards to manage complex internal LLM workflows, boosting operational efficiency across daily research and engineering workflows.
- **Optimized real-time execution visibility:** Designed and integrated a WebSocket-based inference streaming pattern, slashing update latency by **300ms** and delivering immediate visibility into live LLM execution states for prompt optimization.
- **Scaled distributed task processing:** Constructed Apache Airflow DAGs with Celery workers executing 1,000+ batch jobs per worker; implemented custom deduplication logic and comprehensive Jest unit tests to guarantee 99.9% data reliability.
- **Enhanced model response quality:** Integrated OpenAI APIs across enterprise workflows and refined prompt engineering strategies, reducing hallucination rates and increasing output accuracy for technical research teams.

Krishlabs | Junior Full-Stack Developer (Contract) | Bangalore, India |

Nov 2024 – Apr 2025

- **Accelerated customer acquisition:** Developed a responsive Next.js/Express marketing platform with Framer Motion and validated forms, expanding user engagement by **25%** and conversion sign-ups by **40%**.
- **Streamlined real-time communications:** Engineered a cross-platform social mobile application utilizing Expo React Native, Node.js, and MongoDB; integrated Socket.IO real-time channels, reducing messaging latency by **60%**.
- **Deployed resilient cloud infrastructure:** Configured and launched 15+ RESTful API endpoints on AWS EC2 backed by S3 media storage and MongoDB, maintaining **99.8%** service uptime across high-traffic authentication and content services.

OPEN SOURCE CONTRIBUTIONS

- [louislam/uptime-kuma](#) & [lingdojo/kana-dojo](#): Authored dynamic gRPC port binding (PR: fix/grpc-ports) and dashboard filter URL synchronization (+241 lines, PR: feat/sync-filters); contributed minimalist visual UI theme; actively collaborated across 16+ code reviews to elevate test isolation and user experience.

PROJECTS

Gaminai — AI Research Assistant | React • Express.js • TypeScript • Postgres • Prisma ORM | [Live Demo](#)

- **Architected end-to-end RAG architecture:** Built a Retrieval-Augmented Generation pipeline using PostgreSQL, Prisma, and OpenAI/Anthropic APIs with exactly-once processing and real-time streaming for low-latency AI responses.
- **Grounded AI accuracy with real-time web retrieval:** Implemented contextual prompt engineering alongside dynamic web source retrieval, sustaining multi-turn conversation memory and eliminating model hallucinations across user sessions.

ProScope-Focus | YouTube API • React • IndexedDB • Express.js • OAuth 2.0 • Passport.js • Vercel | [Live](#)

- **Optimized offline-first UI responsiveness:** Selected IndexedDB for offline-first state management in React 19, implementing a 600ms debounced sync buffer that constrained local UI updates to under **100ms**.
- **Reduced API rate limits via middleware caching:** Developed an Express.js middleware server with in-memory caching to optimize YouTube metadata retrieval, minimizing external API calls and reducing backend latency.

RunAway | *JavaScript, Canvas 2D, Node.js* | [Live](#) | [Code](#)

- **Built interactive audio/video transport interface:** Integrated YouTube IFrame Player API and Web Audio procedural synthesizer within Canvas 2D for seamless multimedia interaction.
- **Drove organic user adoption:** Attained **30+ daily active users** and **48+ unique visitors** shortly after launch, demonstrating strong user retention and engagement.

ACHIEVEMENTS & CERTIFICATIONS

- **Hackathon:** 3rd place, Unstop Powered Hackathon (200+ participants).
- **Certifications:** JS & Java (IIT Bombay, 90% score). **Ongoing:** DSA practice in Java [GitHub](#).